1. Objective

Create the database and seven tables for the Airline Reservation System using MySQL DDL.

2. SQL Table Creation

```
CREATE DATABASE airline_reservation;
USE airline_reservation;

CREATE TABLE Passenger (
 Passenger_ID INT PRIMARY KEY,
 Name VARCHAR(100) NOT NULL,
 Email VARCHAR(100) UNIQUE NOT NULL,
 Phone VARCHAR(15) NOT NULL,
 Passport_No VARCHAR(20) UNIQUE NOT NULL
);

CREATE TABLE Airport (
 Airport_ID INT PRIMARY KEY,
 Airport_Name VARCHAR(100) NOT NULL,
 City VARCHAR(50) NOT NULL,
 Country VARCHAR(50) NOT NULL,
 Airport_Code CHAR(3) UNIQUE NOT NULL
);

CREATE TABLE Aircraft (
 Aircraft_ID INT PRIMARY KEY,
 Model VARCHAR(50) NOT NULL,
 Capacity INT NOT NULL CHECK (Capacity > 0),
 Manufacturer VARCHAR(50) NOT NULL
);

CREATE TABLE Seat (
 Seat_ID INT PRIMARY KEY,
 Aircraft_ID INT NOT NULL,
 Seat_Number VARCHAR(10) NOT NULL,
 Class VARCHAR(30) NOT NULL DEFAULT 'Economy',
 FOREIGN KEY (Aircraft_ID) REFERENCES Aircraft(Aircraft_ID),
 UNIQUE (Aircraft_ID, Seat_Number),
 CHECK (Class IN ('Economy','Business','First'))
);

CREATE TABLE Flight (
 Flight_ID INT PRIMARY KEY,
 Flight_Number VARCHAR(20) UNIQUE NOT NULL,
 Aircraft_ID INT NOT NULL,
 Departure_Airport INT NOT NULL,
 Arrival_Airport INT NOT NULL,
 Departure_Time DATETIME NOT NULL,
 Arrival_Time DATETIME NOT NULL,
 Status VARCHAR(30) NOT NULL DEFAULT 'Scheduled',
 FOREIGN KEY (Aircraft_ID) REFERENCES Aircraft(Aircraft_ID),
 FOREIGN KEY (Departure_Airport) REFERENCES Airport(Airport_ID),
 FOREIGN KEY (Arrival_Airport) REFERENCES Airport(Airport_ID),
 CHECK (Departure_Airport <> Arrival_Airport),
 CHECK (Arrival_Time > Departure_Time),
 CHECK (Status IN ('Scheduled','Delayed','Departed','Arrived','Cancelled'))
);

CREATE TABLE Booking (
 Booking_ID INT PRIMARY KEY,
 Passenger_ID INT NOT NULL,
 Flight_ID INT NOT NULL,
 Seat_ID INT NOT NULL,
 Booking_Date DATE NOT NULL DEFAULT (CURRENT_DATE),
 Booking_Status VARCHAR(30) NOT NULL DEFAULT 'Confirmed',
 FOREIGN KEY (Passenger_ID) REFERENCES Passenger(Passenger_ID),
 FOREIGN KEY (Flight_ID) REFERENCES Flight(Flight_ID),
 FOREIGN KEY (Seat_ID) REFERENCES Seat(Seat_ID),
 UNIQUE (Flight_ID, Seat_ID),
 CHECK (Booking_Status IN ('Confirmed','Pending','Cancelled','Completed'))
);

CREATE TABLE Payment (
 Payment_ID INT PRIMARY KEY,
 Booking_ID INT NOT NULL UNIQUE,
 Amount DECIMAL(10,2) NOT NULL CHECK (Amount >= 0),
 Payment_Date DATE NOT NULL DEFAULT (CURRENT_DATE),
 Payment_Method VARCHAR(30) NOT NULL,
 Payment_Status VARCHAR(30) NOT NULL DEFAULT 'Pending',
 FOREIGN KEY (Booking_ID) REFERENCES Booking(Booking_ID),
 CHECK (Payment_Method IN ('Card','UPI','Net Banking','Cash')),
 CHECK (Payment_Status IN ('Paid','Pending','Failed','Refunded'))
);
```

3. Relationships

Parent	Child	Relationship
Passenger	Booking	One passenger can make many bookings.
Flight	Booking	One flight can have many bookings.
Aircraft	Flight / Seat	One aircraft can be used for flights and has many seats.
Airport	Flight	Airport is referenced as departure and arrival.
Seat	Booking	Seat allocation is unique per flight.
Booking	Payment	One payment row per booking under the UNIQUE rule.

4. Verify Tables

```
SHOW TABLES;\nDESCRIBE Passenger;\nDESCRIBE Flight;\nDESCRIBE Booking;\nDESCRIBE Payment;
```

Expected tables: Aircraft, Airport, Booking, Flight, Passenger, Payment and Seat.

5. Conclusion

The schema defines the core tables and their relationships. Run the SQL in MySQL Workbench to create the database.